

SATVIK GARG

+1 (604) 764-5434 | sat.garg03@gmail.com | [linkedin.com/in/satvikgarg15](https://www.linkedin.com/in/satvikgarg15) | GitHub: [sgarg15](https://github.com/sgarg15) | Website: sgarg.vercel.app

EDUCATION

Simon Fraser University

Bachelor of Computer Science, AI concentration; Full Ride Scholarship

Vancouver, BC

April 2026

WORK EXPERIENCE

Software Developer Intern

May 2025 – Aug 2025

Amazon

Toronto, ON

- Migrated FNSku Maps backup from Java Reactor to Datapath Precompute, **reducing costs by 40%** and **latency by 20%**
- Authored migration design doc and implemented solution using DQL and CI/CD pipelines with automated testing
- Created bulk FNSku Map upload tool in Java, **automating 300+** requests annually and enabling customer self-service
- Contributed to modernization by migrating additional internal reactors from JDK 8 to JDK 17 for long-term maintainability

Software Undergraduate Research Assistant

May 2024 – Aug 2024

Simon Fraser University

Vancouver, BC

- Conducted research on multi-agent reach-avoid games through simulation and real-world implementations
- Designed Mixed Integer Programming and learning-based control logic to maximize number of attackers captured by defenders and scale to higher-dimensional games
- Controlled TurtleBot3 using ROS and Python for real-life simulations, demonstrating method's effectiveness

Software Quality Assurance Intern

Jan. 2023 – Aug. 2023

The Boeing Company

Richmond, BC

- Developed Java/Selenium E2E and smoke tests, increasing automated regression coverage by **45%**
- Integrated **Adobe Analytics** on multiple application pages, while creating **JUnit** test cases for accurate tracking
- Cooperated closely with infrastructure team to support and manage **CI/CD pipeline** and integrate **E2E tests**
- Achieved significant **40%** reduction in reporting time by automating Excel data collection and visualization of key development metrics using **Power BI** and live data from **Azure DevOps**

PROJECTS

Hamilton-Jacobi UAV Reach-Avoid Game | ROS2, Crazyswarm2, Crazyflie 2.1, Vicon, Python

Jan 2026 – April 2026

- Integrated Crazyflie 2.1 drones with **ROS2/Crazyswarm2** to execute HJ reach-avoid UAV interception on hardware
- Built Vicon motion-capture pipeline for real-time pose tracking, state estimation, and hardware flight validation
- Validated controller across 10 simulated environments, achieving **97%** success rate and **15s** average capture time

Echonote | C, Zephyr RTOS, Microcontroller

Aug 2024 – Nov 2024

- Built a wearable voice-capture prototype on an nRF52840 microcontroller using Zephyr RTOS and BLE
- Implemented BLE communication between embedded firmware and a mobile client for low-power audio/data transfer
- Designed local-first processing and encryption flow to reduce cloud dependency and improve user privacy

SoundSphere | React Native, NodeJS, PostgreSQL, AWS

Dec 2023 – May 2024

- Developed an app that discovers the music taste of others around you by integrating Spotify and Apple Music
- Hosted custom-built REST API on **AWS EC2** instance and the **PostgreSQL** database on **AWS RDS** to ensure scalability and high availability
- Utilized **Redux** and **RTK Query** for efficient state management and data fetching, leading to responsive app performance and user interactions

AWARDS

NSERC USRA Scholarship / Undergrad Research Position in Mars Lab
Simon Fraser University Full-Ride Entrance Scholarship with Distinction
Qualification for the VEX Robotics Competition World Championships

TECHNICAL SKILLS

Languages: C++, C, C#, Python, Java, Dart, JavaScript, TypeScript, CSS, HTML, SQL/NoSQL

Frameworks: React, Angular, Maven, PostgreSQL, Node.js, PyTorch, LangChain, Express.js, Numpy